

Set 1

Section A [2 * 5 = 10 marks]

1. What is the output of the following code?

```
l=['p','r','o','g','r','a','m']
l.remove('p')
print(l)
print(l.pop())
```

2. `print("banana".split('a'))` will result in _____ and `print("abcdef"[-1:-7:-2])` is _____
3. A series of *if-elif* statement have a final _____ block which is executed if none of *if* or *elif* expressions is _____
4. Differentiate break and continue
5. a) Fill in the blank (2+____)**2=25
b) To print "Python is fun", fill in blanks `print("python"+ _____ + _____)`

Section B [4 * 5 = 20 marks] Attempt any 4 programming questions

1	A company pays overtime as follows: First 8 hours → ₹200 per hour Extra hours → ₹300 per hour If salary exceeds ₹5000, deduct 5% tax. Input: Total hours worked Print: Gross salary, tax, net salary. Sample Input 20 Sample Output 4400 0 4400																		
2	Write a Python program to accept marks of 3 subjects and calculate the average and assign grade based on average using below condition : <table border="1"> <thead> <tr> <th>Average</th><th>Grade</th></tr> </thead> <tbody> <tr> <td>90-100</td><td>O</td></tr> <tr> <td>80-89</td><td>A+</td></tr> <tr> <td>70-79</td><td>A</td></tr> <tr> <td>60-69</td><td>B+</td></tr> <tr> <td>50-59</td><td>B</td></tr> <tr> <td>49-45</td><td>C</td></tr> <tr> <td>40-44</td><td>P</td></tr> <tr> <td><40</td><td>F.</td></tr> </tbody> </table> Sample Input: 84 75 88 Output: Average is: 82.33 Grade is A+	Average	Grade	90-100	O	80-89	A+	70-79	A	60-69	B+	50-59	B	49-45	C	40-44	P	<40	F.
Average	Grade																		
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3	Mr. Sumanth is a student of Prof. Suraj Aravind from GITAM Deemed to be University. Mr. Sumanth is assigned a task by his professor to label the numbers starting from X and ending at Y with the following conditions: Condition 1: If the number is a multiple of 3 print "OH" in the place of the number. Condition 2: If the number is a multiple of 5 print "AH" in the place of the number. Condition 3: If the number is a multiple of both 3 & 5 print "OHAH" in the place of the number. Help Sumanth in designing the code to satisfy the given conditions. Sample Input: <pre>3 10</pre> Sample Output: <pre>OH 4 AH OH 7 8 OH AH</pre> Description: X=3 and Y=10, Numbers from X to Y is 3 4 5 6 7 8 9 10, Since 3,6,9 is the multiple of 3 in the place of that numbers OH is printed. 5 and 10 are the multiple of 5 in the place of that numbers AH is printed. No numbers in the range which are multiple of 3 and 5.
4	Given a string. Cut it into two "equal" parts (If the length of the string is odd, place the center character in the first string, so that the first string contains one more character than the second). Now print a new string on a single row with the first and second halves interchanged (second half first and the first half second) Don't use the <i>if</i> statement in this task. Sample Input :Hello Sample Output: loHel Sample Input :Qwerty Sample Output: rtyQwe
5	A company stores employee details in a nested list. Each employee record contains: [id, name, dob (dd-mm-yyyy), salary, department] The HR manager wants to identify all employees who work in the Sales department for a performance review. Also print the highest paid employee Input: <pre>[[1, 'lalith', '12-12-2000', 25000, 'banking'], [2, 'raju', '02-05-2001', 50000, 'sales'], [3, 'sita', '15-08-1999', 45000, 'sales']]</pre> Output:Sales: <pre>raju sita Highest paid:sita</pre>
6	A college management system stores student names in two different data structures: • The first names of students are stored in a list • The last names of students are stored in a tuple Write a python program to print the names of the students, Example : Sample Input: <pre>first_names = ["Anil", "Priya", "Rahul", "Sneha"] last_names = ("Kumar", "Reddy", "Sharma", "Verma")</pre> Output: ["Anil Kumar", "Priya Reddy", "Rahul Sharma", "Sneha Verma"]

Set 2**Section A [2 * 5 = 10 marks]**

1. What is output of

```
x = 6
```

```
y = 2
```

```
print(x ** y)
```

```
print(x // y)
```

2. How to check whether a string contains only digits and whether it contains only alphabets?

3. value = 4 Fill in code such that value becomes 12 value _____ 3
and 10 _____ 3 gives 3

4. Write code to print pattern

```
*  
  
* *  
  
* * *  
  
* * * *
```

5. `l=[10,20,30,40,50]`

```
l1=[i if i%4==0 for i in l]
```

```
print(l1)
```

Section B [4 * 5 = 20 marks] Attempt any 4 programming questions

1	Shanmukesh has M cookies and Gurupreet has N cookies. They are playing a game by exchanging their cookies with each other. (Shanmukesh gives all his cookies to Gurupreet and viceversa). Write a program in Python to display how many cookies are there with Shanmukesh and Gurupreet after their 1st exchange. Input: 19 21 Output: Cookies at the beginning: 19, 21 Cookies after 1st interchange: 21, 19
2	Saatvik measures three sides of a triangle at a construction site. He wants to check whether it is a right-angled triangle using the Pythagoras theorem. Write a Python program that takes three side lengths as input and prints whether the triangle is right-angled or not right-angled. Input: 3,4,5 Output: Yes Input: 4,5,6 Output: No

3	Help Mr. Pardhiv in the Design of Automated Machine to print the series of Fibonacci Numbers by helping him to understand the concept of the Fibonacci Number. Fibonacci Series is Generated by taking the first number of the series as always 0 and the second number of the series as always 1, the next number of the series has been computed by adding the previous two numbers of the series. The Designed Automated Machine has to print the first n numbers in the Fibonacci. Sample Input: 5 Sample Output: 0 1 1 2 3
4	Imagine a signal tower sending a long sequence of signals, represented by a string S. Some signals are sent again and again (repeating characters). Some signals are sent only once. The control room wants to find the first unique signal—the one that appeared once and never repeated. No unique signal detected → print -1 Input: S = "radar" Output: 'd' Input: S="mallam" Output: -1
5	A teacher has two lists of student roll numbers. He wants to check whether both lists are exactly the same (same elements in the same order) Write a Python program that takes two lists as input and prints whether the lists are equal or not equal. List 1: [1, 2, 3, 4] List 2: [1, 2, 3, 4] Output: Lists are equal List 1: [1, 2, 3, 4] List 2: [4, 3, 2, 1] Output: Lists are not equal
6	Given a tuple of strings apply an encoding technique on each string and print the tuple as shown in the example below Example Input : ("aaabbbc","ssrrd","abc") Output : ("a3b2c1","s2r2d1","a1b1c1")

Set 3

Section A [2 * 5 = 10 marks]

1. To print the original string in reverse order we can set the step to _____
Convert string s to lowercase() using _____
2. What is output of following

<pre>l=[1,2,3] for i in l: l.append(i) if len(l)>5: break print(l) ans:_____ _____</pre>	<pre>D=[1,2,[1,2],7] print(D[2]) ans_____</pre>
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3. Differentiate List and Tuple
4. What is $2^{**}3^{**}2$ _____ and $10+4*3-6/2$ _____
5. Fill in code to print even numbers in range a to b
for i _____:
 if _____:
 print(i)

Section B [4 * 5 = 20 marks] Attempt any 4 programming questions

1	A bakery sells 3 types of cakes. If the total bill exceeds ₹600, a 10% discount is given. Otherwise, no discount. Input: Prices of 3 cakes Quantities of 3 cakes Print: Original total, discount amount, final amount. Sample Input: 100 150 200 2 1 2 Sample Output: 750 75.0 675.0
2	Write a python program to check if a given year is leap or not. A year is a leap year if it is divisible by 4 and not divisible by 100, or if it is divisible by 400. Test case 1: Input: 2024 Output: Leap Test Case 2: Input: 1900 Output: Not a Leap year (reason divisible by 4 but also divisible 100).

3	In the Number Theory, the Professor assigns a task to Mr. Saran Teja to find the factors of a given number. Mr. Saran Teja has understood the concept of factors and is able to find all the factors of the input number. Later Mr. Saran Teja challenged himself to identify only the even factors for the input number. Design a code to help Mr. Saran Teja to find the Even factors of the input numbers. Sample Input: 6 Sample Output: The Even factors of 6 is: 2 6
4	A password strength checker validates passwords with these requirements: Password must contain at least one uppercase letter, Password must contain at least one lowercase letter Password must contain at least one digit, Password must contain at least one special character (!@#\$%^&*) Password length must be between 8 and 20 characters Calculate strength score: weak (1-2 criteria), medium (3 criteria), strong (4-5 criteria) Task: Write a Python program to validate and rate password strength. Input: MyP@ss123 Output: Valid Password: Yes Strength: Strong Criteria Met: 5/5 Length: 9
5	A teacher conducted a class survey asking students to choose their favorite programming language. The responses were stored in a list. The teacher wants to know how many times each programming language was chosen. Write a Python program that: Takes a list of elements as input. Counts the frequency of each element in the list. Displays each element along with its frequency. Input: Enter elements: ['Python', 'Java', 'Python', 'C', 'Java', 'Python'] Output: Python : 3 Java : 2 C : 1
6	A company maintains employee records in the form of a tuple containing the following details: (eid, name, salary, designation) The management has decided to revise employee salaries based on the following criteria: If the salary is greater than ₹50,000, provide a 10% hike. If the salary is between ₹30,000 and ₹50,000 (inclusive), provide an 8% hike. If the salary is less than ₹30,000, provide a 5% hike. Write a Python program to read details of n employees and store them as tuples. Apply the appropriate salary hike based on the given conditions. Update the salary accordingly. Display the updated employee records. Example: Input: Enter number of employees: 2 Employee 1: (101, "Ravi", 55000, "Manager") Employee 2: (102, "Priya", 28000, "Developer") Output: (101, "Ravi", 60500, "Manager") (102, "Priya", 29400, "Developer")

Set 4

Section A [2 * 5 = 10 marks]

- a) What is output of `print("mississippis".replace("ss","X"))` _____

b) Output of `"hello".title()` is _____
- Boolean values in python are _____ and _____
- What is the output of

```
l=[1,2,3,4,5]
print(l[1:-1])
print(l[-3:-1])
```
- `not 1=1` gives _____ and python uses _____ to form a block of code
- To specify an empty body of a for loop, use _____ statement.

Section B [4 * 5 = 20 marks] Attempt any 4 programming questions

1	A ride-sharing app calculates fare as: Base Fare = ₹50 ₹12 per kilometer If ride distance > 15 km, add 20% surge charge Write a program to calculate the total fare. Input: 18 km Output: Total Fare = 309.6
2	Write a Python program to find the absolute value of a number. The program should convert negative numbers to positive while keeping positive numbers unchanged. Input:-8 Output:8 Input:5 Output:5
3	At University, three teams participated in a coding hackathon. Each team completed the challenge in different amount of time (in minutes). The event coordinator wants to identify which team finished in the least amount of time, so that the fastest team can be awarded. You are assigned to write a Python program to help determine the minimum time among the three teams. Input: Enter time of Team 1: 120 Enter time of Team 2: 95 Enter time of Team 3: 110 Output: Fastest completion time is: 95 minutes

4	A school is organizing students for the Annual Day group photo. The teacher decides to arrange students in a special triangular formation: Row 1 → 1 student Row 2 → 2 students Row 3 → 3 students Row 4 → 4 students and so on... Each student is given a consecutive roll number starting from 1. Your task is to print the arrangement of roll numbers for N rows. This arrangement forms a pattern called Floyd's Triangle. Write a program to print the numbers in the form of Floyd's triangle. Example: Input: 4 Output: 1 2 3 4 5 6 7 8 9 10
5	Write a Python program that: • Removes any leading or trailing spaces from the file name. • Checks whether the file name and extension are separated by a dot (.). • Validate that the file name contains only alphanumeric characters. • Checks whether the extension belongs to the .pdf/.txt/.doc/.py list. • Displays whether the file name is Valid or Invalid. Input: gitam2026.pdf Output: Valid File"
6	A college maintains a list of student IDs who attended a workshop. Due to a system error, some student IDs were recorded multiple times. The administrator wants to remove duplicate IDs but maintain the original order of attendance. input:[101, 102, 103, 101, 104, 102, 105] output:[101, 102, 103, 104, 105]

Set 5

Section A [2 * 5 = 10 marks]

1. Given (*subject* = "Programming in Python"),
What will *subject.title()* return? _____
What will *subject.find("Python")* return? _____
2. What will be the output of the following?

```
L=[0,1,2,3,4]
print(L[-1:-4:-1]) _____
print(L==L[:], Lis L[:]) _____
```
3. A string is appended to another string by _____ operator and repeated by _____ operator.
4. *insert()*, *sort()* returns _____, _____
0. Fill in code to print factors of n

```
i=_____
while i<=n:
    if _____:
        print(i)
```

Section B [4 * 5 = 20 marks] Attempt any 4 programming questions

1	A stationery shop gives 10% discount if the total bill exceeds ₹500 otherwise 0% discount. Input: price for 3 items and quantity for 3 items. Compute total price and then apply discount if eligible. Print original total, discount amount, and final amount to pay. Input: 50 40 80 4 3 6 Output: 800 80 720 Input: 30 55 40 3 4 4 Output: 470 0 470
2	Given three numbers, return true if they are evenly spaced; that is, sorted in ascending or descending order, with the same distance between them Input: 7 9 12 Output: True Input: 30 20 10 Output: True Input 3 9 5 Output: False
3	A team of archaeologists discovers an ancient treasure chest in a temple. The chest has a digital lock that opens only when a Perfect Number is entered. The inscription on the chest reads: "The key is a number equal to the sum of its proper divisors (excluding the number itself). Only then shall the treasure be revealed." You are given a positive integer N. Your task is to help the archaeologists determine whether the number is a Perfect Number.

4	A secure communication system transmits messages as strings. For security reasons, the system applies a multi-rule scrubbing algorithm to remove suspicious characters. Each character in the message is indexed starting from 0. The scrubbing rules are: 1. Delete every character whose index is divisible by 3 2. From the remaining characters: Delete any digit (0–9) that appears at an even index 3. From the final string: If the same character appears consecutively, keep only one occurrence. If after applying all rules the message becomes empty, output -1. Your task is to generate the final decrypted message. Example1: Input: a1b2c3d4e5f6 Output: Bcdef Input: aa11bb22cc Output: B Input: 123456789 Output: -1
5	Given a list of integers, return the average of all elements that are not the maximum or minimum. Assume that there is only one instance of each of the maximum or minimum. There will be at least 3 elements in the given list. Input: [1,2,3,4,5,6] output:3.5 Input: [5,6,1] output:5.0
6	Given a tuple of student marks in five subjects print the count of students who are in different grades. avg>90 O grade avg 80-90 A grade avg 70-80 B grade avg <70 c grade Input: ([95, 92, 90, 93, 96], [85, 80, 82, 88, 90], [72, 75, 70, 78, 74], [60, 65, 68, 62, 66]) Output: O Grade: 1 A Grade: 1 B Grade: 1 C Grade: 1